Namal University Mianwali

Digital Logic Design

Electronic Voting Machine

Name	Registration Number
Muhammad Wajid Jamal	NUM-BSEE-2022-24
Muhammad Usman Shah	NUM-BSEE-2022-28
Ghulum Muhammad Khan	NUM-BSEE-2022-35

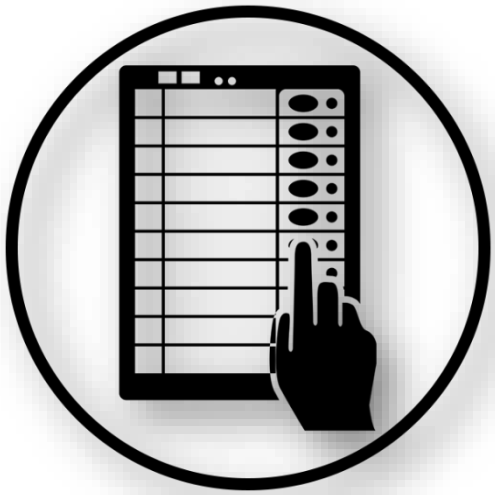


Table of Contents

Abstract: - 3

Introduction: - 4

Literature Existence: - 4

Methodology: - 4

Security System: 4

EVM Counter: 4

4-bit Comparator: 4

Circuit Diagram in Logisim: - 5

Description of the Circuit: - 5

Circuit: - 6

Explanation: - 6

Verilog Code: - 6

Hardware: - 9

Result and Discussion: - 9

Logisim Simulation: - 9

Verilog Implementation: - 9

Hardware Results: 10

Discussion: 10

Future recommendations: - 10

Conclusion: - 10

References:- 11

Abstract: -

The Electronic Voting Machine (EVM) project introduces a secure voting system with three key components. The Security System employs a unique binary password concept, requiring users to input a specific binary value (equivalent to 15 in decimal) to commence voting. The EVM Counter efficiently tallies votes for two candidates, displaying results on dual screens. A 4-bit Comparator then determines the winner. Utilizing the Verilog language, the system undergoes thorough testing,

Introduction: -

The Electronic Voting Machine (EVM) project is all about making voting safer and easier. We made three important parts. First, there's a special Security System. People need to put in a secret binary password to start voting – it's like a secret code to keep things safe.

Then, there's the EVM Counter. It counts votes for two candidates and shows the results on two screens. It's like a helper that makes sure the votes are counted right.

Lastly, there's the 4-bit Comparator. It decides the winner by comparing the votes. We used a computer language called Verilog to make everything work together. After lots of testing, we're confident this project is a good move to make voting simple and better for everyone.

Here's a simplified version:

Literature Existence: -

When we create Electronic Voting Machines (EVMs), we want to make sure voting is safe. One cool idea we found in our readings is called the "Existence Method." It's like a double-check for security – users have to enter both numbers and passwords using 8 buttons in our project. We use XNOR and AND gates for extra safety. To keep things easy, we use 0s and 1s for numbers and passwords. This helps with a 4-bit counter to make sure only the right people can vote. The literature says using clever methods like the Existence Method is key to making EVMs that are both smart and trustworthy.

Methodology: -

The project has three main parts:

Security System:

- Uses 8 buttons (4 for existing numbers, 4 for user passwords).
- Use XNOR and AND gates to keep things safe.
- Connects to a 4-bit counter that lets people start voting.

EVM Counter:

- Uses two 4-bit counters to count votes for two candidates.
- Connected to a common JK flip-flop to count votes together.
- Shows results on two screens with 7-segment displays.

4-bit Comparator:

- Checks who got more votes between the two candidates.
- Uses XNOR and AND gates for the comparison.
- Tells us which candidate has more votes in the end.

Block Diagram

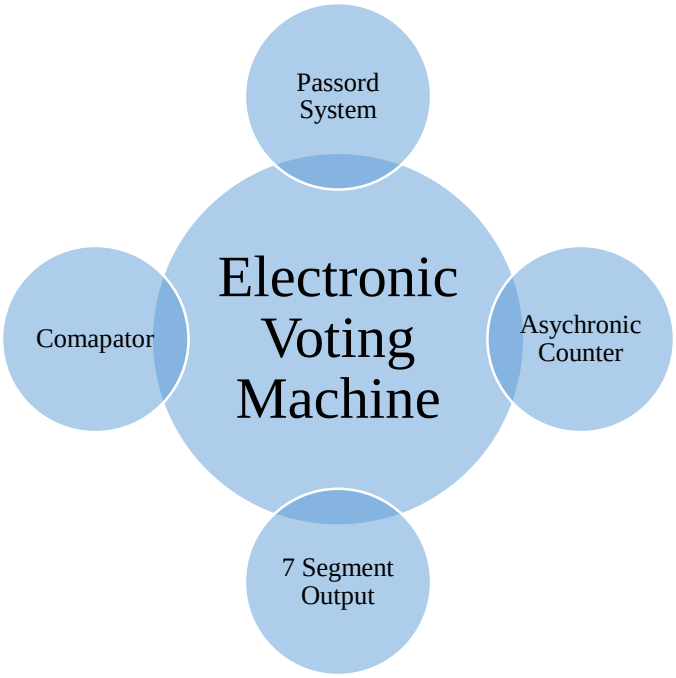


Figure 1 Block Diagram

Circuit Diagram in Logisim: -

To visually represent the hardware design, a detailed circuit diagram was created using the Logisim simulation tool. The diagram showcases the interconnected components of the Electronic Voting Machine (EVM) project, illustrating the logical flow and relationships between the Security System, EVM Counter, and 4-bit Comparator.

Description of the Circuit: -

The circuit begins with the integration of 8 buttons into the Security System, with 4 allocated for existing numbers and 4 for user passwords. XNOR gates are strategically implemented to enforce the dual-layer verification process, ensuring secure user authentication. Binary representation is employed to simplify and secure the input process.

Circuit: -

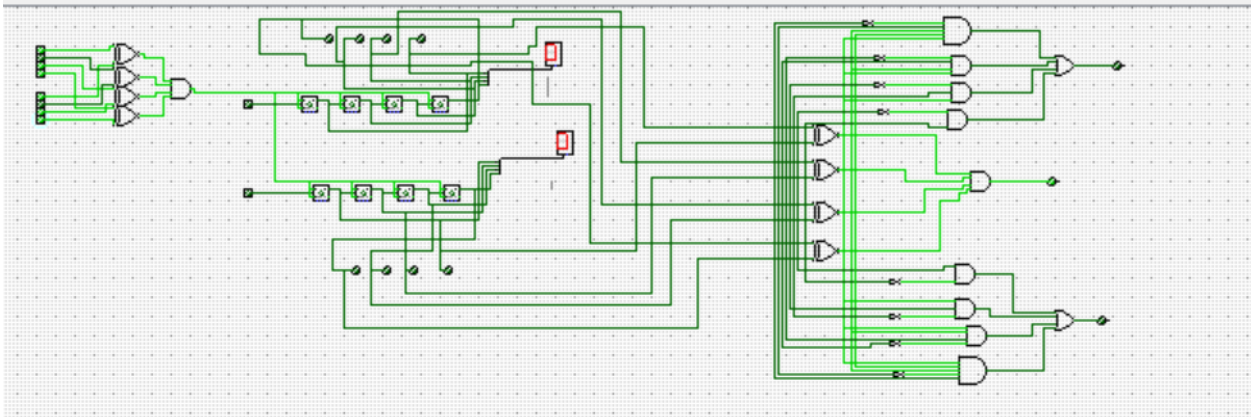


Figure 2Circuit Diagram in Logisim

Explanation: -

- Each component is visually represented, showcasing their connections and interactions within the EVM system.
- Clear labels and symbols are used to denote buttons, XNOR gates, counters, displays, and other essential elements.
- The diagram serves as a comprehensive overview of the logical structure and functionality of the EVM.

Verilog Code: -

```
module up_counter(input clk, reset, output[3:0] counter
    );
reg [3:0] counter_up;

// up counter
always @(posedge clk or posedge reset)
begin
if(reset)
 counter_up <= 4'd0;
else
 counter_up <= counter_up + 4'd1;
end
assign counter = counter_up;
endmodule
module upcounter_testbench();
reg clk, reset;
wire [3:0] counter;

up_counter dut(clk, reset, counter);
initial begin
clk=0;
forever #5 clk=~clk;
end
initial begin
```

```
reset=1;
#20;
reset=0;
end
endmodule
```

Output:-



Figure 3 Verilog Output

Comparator:

Code:

```
module comparator(a,b,eq,lt,gt);

input [3:0] a,b;

output reg eq,lt,gt;

always @(a,b)
begin
 if (a==b)
 begin
  eq = 1'b1;
  lt = 1'b0;
  gt = 1'b0;
 end
 else if (a>b)
 begin
  eq = 1'b0;
  lt = 1'b0;
  gt = 1'b1;
 end
 else
```

```
 begin
  eq = 1'b0;
  lt = 1'b1;
  gt = 1'b0;
 end
end
endmodule
module comparator_tst;
  reg [3:0] a,b;
  wire eq,lt,gt;

  comparator DUT (a,b,eq,lt,gt);

  initial
  begin
   a = 4'b1100;
   b = 4'b1100;
   #10;

   a = 4'b0100;
   b = 4'b1100;
   #10;

   a = 4'b1111;
   b = 4'b1100;
   #10;
   a = 4'b0000;
   b = 4'b0000;
   #10;
   $stop;
  end
endmodule
```

Output:

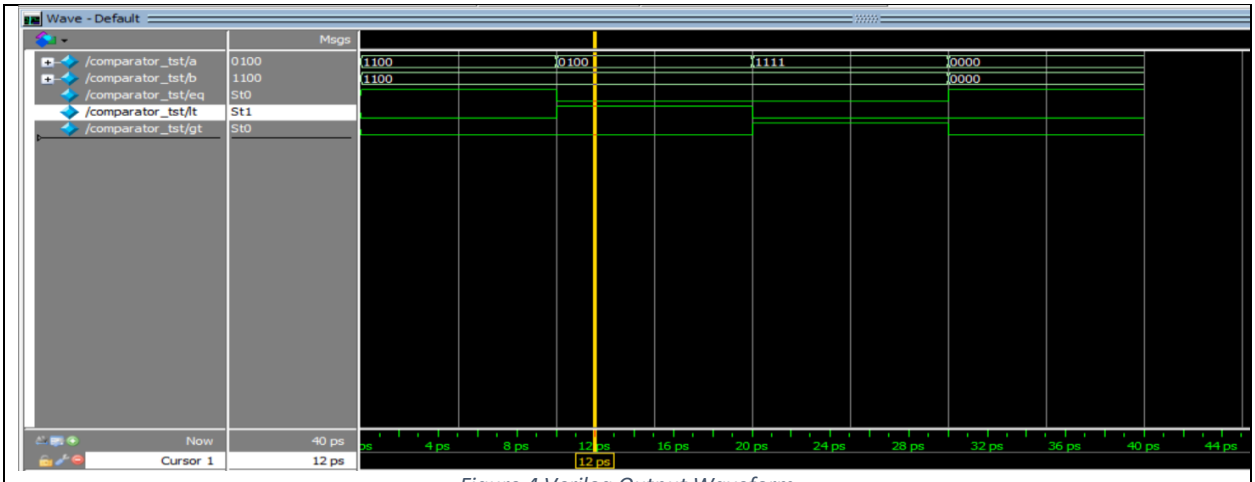


Figure 4 Verilog Output Waveform

Hardware: -

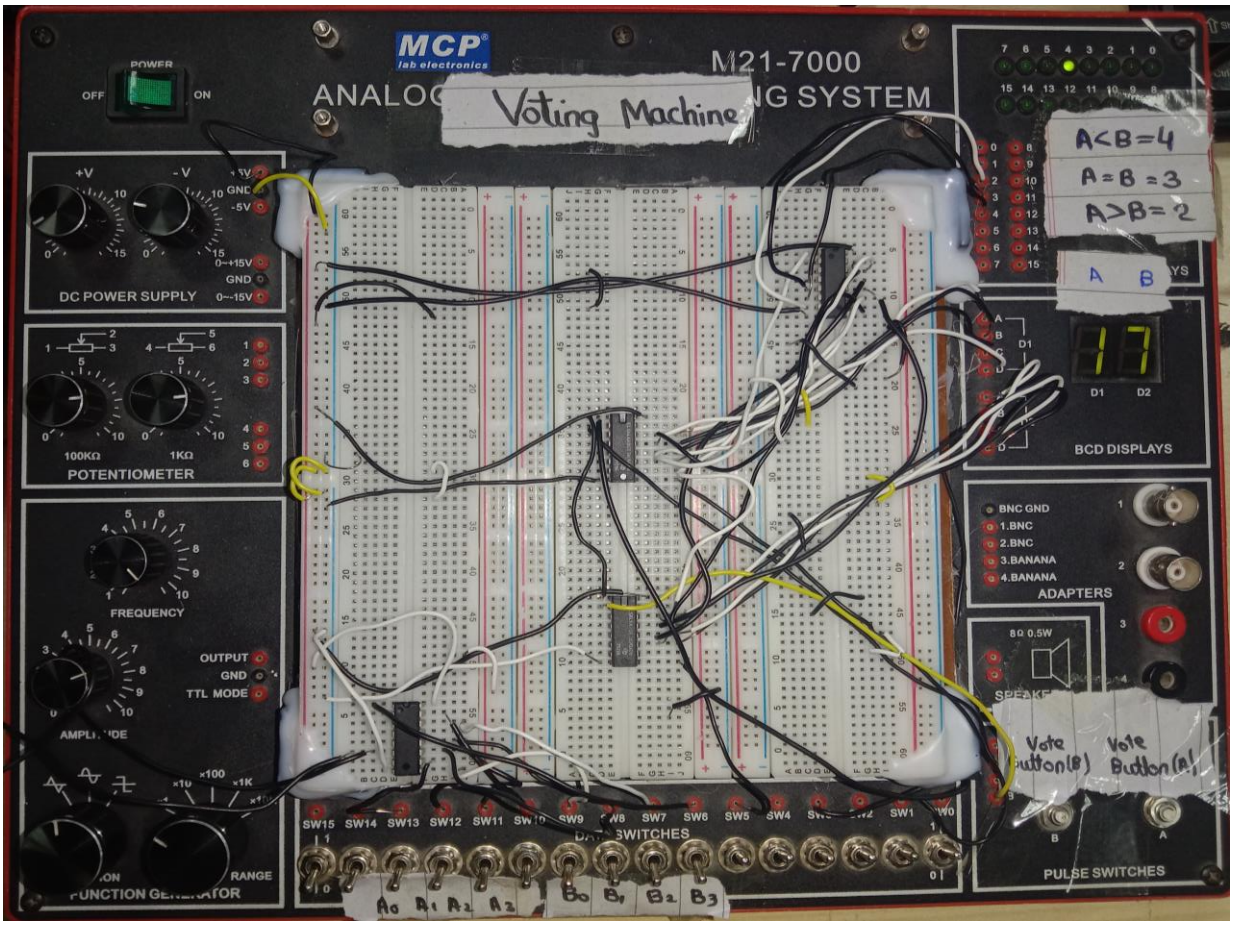


Figure 5 Hardware Implementation

Result and Discussion: -

Logisim Simulation: -

We started by using Logisim to simulate our Electronic Voting Machine (EVM) design. This helped us see how our system would work without building the actual hardware. In Logisim, we could test different scenarios and check if the buttons, gates, and counters behaved as we planned. It was like a practice run before going to the real hardware.

Verilog Implementation: -

Once we were confident with the Logisim simulation, we moved on to using Verilog. Verilog is like the language that tells the hardware what to do. We translated our design into Verilog code to make the real hardware work just like we tested in Logisim. This step was crucial because it turned our plans into something the actual Electronic Voting Machine could understand and follow.

Hardware Results:
Then came the exciting part – building the actual Electronic Voting Machine! We connected all the buttons, gates, and counters, just like in our Logisim simulation and Verilog code. When we tested the real hardware, it matched what we saw in Logisim and what we wrote in Verilog. The buttons worked, the counters counted votes, and the displays showed the results. This showed us that our design, tested in Logisim and translated into Verilog, successfully became a real, working Electronic Voting Machine.

Discussion:
Using Logisim allowed us to play around with our design without worrying about breaking anything. Verilog helped us turn our design into something the hardware could understand. When we put it all together, the hardware matched what we planned and tested. This means our Electronic Voting Machine is reliable and does what it's supposed to do. The combination of Logisim, Verilog, and actual hardware testing was key to making sure our EVM works smoothly and securely.

This journey from simulation to Verilog and finally to real hardware gives us confidence that our Electronic Voting Machine is ready for secure and efficient voting in real-world scenarios.

Future recommendations: -

Looking ahead, our Electronic Voting Machine (EVM) could see improvements in user-friendliness by adopting simpler buttons or touchscreens. Enhancing security with features like biometrics and exploring accessibility options, such as audio instructions, would make the EVM more inclusive. Considering the potential for remote voting through secure online platforms and implementing regular software updates will keep the system technologically current and adaptable. Emphasizing community feedback and ongoing improvements ensures that the EVM remains a reliable and user-friendly tool for future elections.

Conclusion: -

Our Electronic Voting Machine (EVM) project turned out pretty great. We made a safe and smooth system using Logisim, Verilog, and testing it out with real hardware. Going forward, we're thinking about making it even easier for everyone to vote. Like using simpler buttons, adding more security, and making sure everyone can use it without any trouble. We'll keep updating it regularly and listening to what people have to say. So, our EVM is ready to make voting super easy for everyone.

References:-

[1] (PDF) A Simplified Electronic Voting Machine System (researchgate.net)

[2] TRESemme - Pro On The Go (youtube.com)

[3] What are logic gates? | Definition from TechTarget